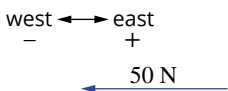


Chapter 6 Scalars and vectors

Section 6.1 Scalars and vectors

Worked example: Try yourself 6.1.1

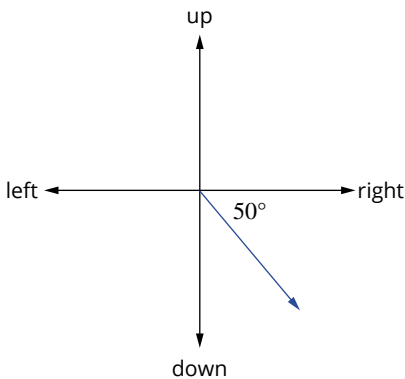
DESCRIBING VECTORS IN ONE DIMENSION

 Describe the vector using:	
a the direction convention shown	
Thinking	Working
Identify the direction convention being used in the vector.	In this case the vector is pointing to the west according to the direction convention.
Note the magnitude, unit and direction of the vector.	In this example the vector is 50 N west.

b an appropriate sign convention.	
Thinking	Working
Convert the physical direction to the corresponding mathematical sign.	The direction west is negative.
Represent the vector with a mathematical sign, magnitude and unit.	This vector is -50 N .

Worked example: Try yourself 6.1.2

DESCRIBING TWO-DIMENSIONAL VECTORS

Describe the direction of the following vector using an appropriate method. 	
Thinking	Working
Choose the appropriate points to reference the direction of the vector. In this case using the horizontal reference makes more sense, as the angle is given from the horizontal.	The vector can be referenced to the horizontal.
Determine the angle between the reference direction and the vector.	From the right direction to the vector there is an angle of 50° .
Determine the direction of the vector from the reference direction.	From the right direction, the vector is down.
Describe the vector using the sequence: angle, clockwise or anticlockwise from the reference direction.	This vector is 50° down from horizontal to the right.

Section 6.1 Review

KEY QUESTIONS SOLUTIONS

- Scalar measures require a magnitude (size) and units.
- Vectors require a magnitude, units and a direction.

Scalar	Vector
time	force
distance	acceleration
volume	position
speed	displacement
temperature	momentum
	velocity

- If the shortest arrow is 2.7 N, the middle length arrow is twice the length of the shortest (5.4 N) and the longest is three times the shortest (8.1 N). The 9.0 N magnitude is not required.
- If the shortest arrow is -5.4 N, the middle length arrow is twice the length of the shortest (10.8 N) and the longest is three times the shortest (16.2 N). The -2.7 N magnitude is not required.
- down
 - south
 - forwards
 - up
 - east
 - positive
- Terms like north and left cannot be used in a calculation. + and - can be used to do calculations with vectors.
- The vector diagram shows -35 N.
- 225° T
 - S 45° W
 - 120° T
 - S 60° E
- 40° up from horizontal to the left

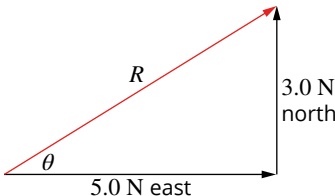
Section 6.2 Adding vectors in one and two dimensions

Worked example: Try yourself 6.2.1

ADDING VECTORS IN ONE DIMENSION USING ALGEBRA

Use the sign and direction conventions shown in Figure 6.2.2 to determine the resultant force on a box that has the following forces acting on it: 16 N up, 22 N down, 4 N up and 17 N down.	
Thinking	Working
Apply the sign and direction conventions to change the directions to signs.	16 N up = +16 N 22 N down = -22 N 4 N up = +4 N 17 N down = -17 N
Add the magnitudes and their signs together.	Resultant force = (+16) + (-22) + (+4) + (-17) = -19 N
Refer to the sign and direction convention to determine the direction of the resultant force vector.	Negative is down. ∴ Resultant force = 19 N down

Worked example: Try yourself 6.2.2
ADDING VECTORS IN TWO DIMENSIONS USING GEOMETRY

Determine the resultant force when forces of 5.0 N east and 3.0 N north act on a tree. Refer to Figure 6.2.2 on page 181 for sign and direction conventions if required.	
Thinking	Working
Construct a vector diagram showing the vectors drawn head to tail. Draw the resultant vector from the tail of the first vector to the head of the last vector.	
As the two vectors to be added are at 90° to each other, apply Pythagoras' theorem to calculate the magnitude of the resultant vector.	$R^2 = 5.0^2 + 3.0^2$ $= 25 + 9$ $R = \sqrt{34}$ $= 5.8 \text{ N}$
Using trigonometry, calculate the angle from the east vector to the resultant vector.	$\tan \theta = \frac{3.0}{5.0}$ $\theta = \tan^{-1} 0.6$ $= 31.0^\circ$
Determine the direction of the vector relative to north or south.	$90^\circ - 31^\circ = 59^\circ$ The direction is N 59° E
State the magnitude and direction of the resultant vector.	$R = 5.8 \text{ N, N } 59^\circ \text{ E}$

Section 6.2 Review

KEY QUESTIONS SOLUTIONS

- Total weekly distance = $5 \times 2 \times 3.0 = 30 \text{ km}$
 - Since he returns home each day, his displacement is zero each day and each week.
- Using sign conventions, resultant = $+3 - 2 - 3 = -2$. The resultant vector is 2 m down.
- Using sign conventions, resultant = $+23 + (-16) + 7 + (-3) = +11$. The resultant vector is 11 m forwards.
- D. Adding vector B to vector A is equivalent to saying $A + B$. Therefore, draw vector A first, then draw vector B with its tail at the head of A. The resultant is drawn from the tail of the first vector (A) to the head of the last vector (B).
- $$R^2 = 40.0^2 + 20.0^2$$

$$= 1600 + 400$$

$$R = \sqrt{2000}$$

$$= 44.7 \text{ m}$$

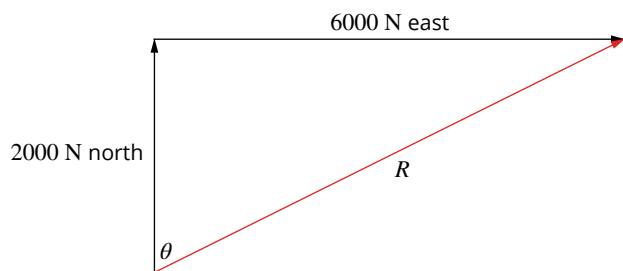
$$\tan \theta = \frac{40.0}{20.0}$$

$$\theta = \tan^{-1} 2.00$$

$$= 63.4^\circ$$

$$R = 44.7 \text{ m, S } 63.4^\circ \text{ W}$$

6



$$\begin{aligned} R^2 &= 2000^2 + 6000^2 \\ &= 4\,000\,000 + 36\,000\,000 \\ R &= \sqrt{40\,000\,000} \\ &= 6325 \text{ N} \end{aligned}$$

$$\begin{aligned} \tan \theta &= \frac{6000}{2000} \\ \theta &= \tan^{-1} 3.00 \\ &= 71.6^\circ \\ R &= 6325 \text{ N, N } 71.6^\circ \text{ E} \end{aligned}$$

7

$$\begin{aligned} R^2 &= 40.0^2 + 30.0^2 \\ &= 1600 + 900 \\ R &= \sqrt{2500} \\ &= 50.0 \text{ m} \end{aligned}$$

8 First add 3000 N north and 5000 N south
Resultant force is 2000 N south
Then add 2000 N south to 5000 N east

$$\begin{aligned} F^2 &= 2000^2 + 5000^2 \\ &= 29\,000\,000 \\ F &= 5385 \text{ N} \\ \theta &= \tan^{-1} \frac{5000}{2000} = 68.2^\circ \end{aligned}$$

Resultant force = 5385 N S 68.2° E

9 Total forwards force = 3350 + 2235 + 634 = 6219 N
Apply a sign convention: forwards = + 6219 N; backward = -6220 N
Add vectors = +6219 - 6220 N = -1 N
Resultant force = 1 N backwards

Section 6.3 Subtracting vectors in one and two dimensions

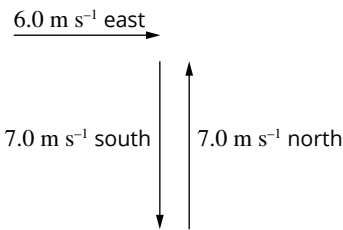
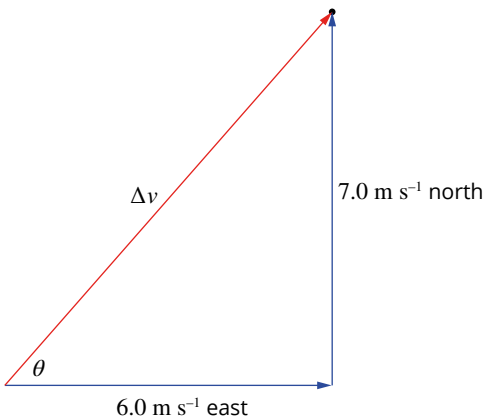
Worked example: Try yourself 6.3.1

SUBTRACTING VECTORS IN ONE DIMENSION USING ALGEBRA

Use the sign and direction conventions shown in Figure 6.3.5 on page 188 of the Student Book to determine the change in velocity of a rocket as it changes from 212 m s^{-1} up to 2200 m s^{-1} up.	
Thinking	Working
Apply the sign and direction conventions to change the directions to signs.	$v_1 = 212 \text{ m s}^{-1} \text{ up} = +212 \text{ m s}^{-1}$ $v_2 = 2200 \text{ m s}^{-1} \text{ up} = +2200 \text{ m s}^{-1}$
Reverse the direction of the initial velocity, v_1 , by reversing the sign.	$-v_1 = 212 \text{ m s}^{-1} \text{ down}$ $= -212 \text{ m s}^{-1}$
Use the formula for change in velocity to calculate the magnitude and the sign of Δv .	$\Delta v = v_2 + (-v_1)$ $= +2200 + (-212)$ $= +1988 \text{ m s}^{-1}$
Refer to the sign and direction convention to determine the direction of the change in velocity.	Positive is up $\therefore \Delta v = 1988 \text{ m s}^{-1} \text{ up}$

Worked example: Try yourself 6.3.2

SUBTRACTING VECTORS IN TWO DIMENSIONS USING GEOMETRY

Determine the change in velocity of a ball as it bounces off a wall. The ball approaches at 7.0 m s^{-1} south and rebounds at 6.0 m s^{-1} east.	
Thinking	Working
Draw the final velocity vector, v_2 , and the initial velocity vector, v_1 , separately. Then draw the initial velocity in the opposite direction.	
Construct a vector diagram, drawing v_2 first, and then from its head draw the opposite of v_1 . The change in velocity vector is drawn from the tail of the final velocity to the head of the opposite of the initial velocity.	
As the two vectors to be added are at 90° to each other, apply Pythagoras' theorem to calculate the magnitude of the change in velocity.	$R^2 = 7.0^2 + 6.0^2$ $= 49 + 36$ $R = \sqrt{85}$ $= 9.2 \text{ m s}^{-1}$

Calculate the angle from the north vector to the change in velocity vector.	$\tan \theta = \frac{7.0}{6.0}$ $\theta = \tan^{-1} 1.17$ $= 49.40^\circ$ Direction from north vector is $90 - 49.40 = 40.60^\circ$
State the magnitude and direction of the change in velocity.	$\Delta v = 9.2 \text{ ms}^{-1} \text{ N } 41^\circ \text{ E}$

Section 6.3 Review

KEY QUESTIONS SOLUTIONS

- Change in velocity = final velocity – initial velocity
 $= 5 + (+3)$
 $= 8 \text{ ms}^{-1} \text{ east}$
- Change in velocity = final velocity – initial velocity
 $= 2 + (-4)$
 $= 2 \text{ ms}^{-1} \text{ left}$
- Change in velocity = final velocity – initial velocity
 $= -3 + (-4)$
 $= 7 \text{ ms}^{-1} \text{ downwards}$
- Change in velocity = final velocity – initial velocity
 $= -32.5 + (-35.0)$
 $= 67.5 \text{ ms}^{-1} \text{ south}$
- Change in velocity = final velocity – initial velocity
 $= 8.2 + (-22.2)$
 $= 14.0 \text{ ms}^{-1} \text{ backwards}$
- $\Delta v^2 = (v_2)^2 + (-v_1)^2$
 $= (406)^2 + (345)^2$
 $\Delta v = \sqrt{1648.36 + 1190.25}$
 $= \sqrt{2838.61}$
 $= 533 \text{ ms}^{-1}$
 $\tan \theta = \frac{345}{406}$
 $\theta = \tan^{-1} \frac{345}{406}$
 $= 40.4^\circ$
 Angle measured from the north $= 90^\circ - 40.4^\circ = 49.6^\circ$
 $\Delta v = 533 \text{ ms}^{-1} \text{ N } 49.6^\circ \text{ W}$
- $\Delta v^2 = (v_2)^2 + (-v_1)^2$
 $= (42.0)^2 + (42.0)^2$
 $\Delta v = \sqrt{1764 + 1764}$
 $= \sqrt{3528}$
 $= 59.4 \text{ ms}^{-1}$
 $\tan \theta = \frac{42.0}{42.0}$
 $\theta = \tan^{-1} (1.000)$
 $= 45.0^\circ$
 $\Delta v = 59.4 \text{ ms}^{-1} \text{ N } 45.0^\circ \text{ W}$

$$\begin{aligned}
 8 \quad \Delta v^2 &= (v_2)^2 + (-v_1)^2 \\
 &= (5.25)^2 + (7.05)^2 \\
 \Delta v &= \sqrt{(27.56 + 49.70)} \\
 &= \sqrt{77.27} \\
 &= 8.79 \text{ m s}^{-1}
 \end{aligned}$$

$$\tan \theta = \frac{7.05}{5.25}$$

$$\begin{aligned}
 \theta &= \tan^{-1} \frac{7.05}{5.25} \\
 &= 53.3^\circ
 \end{aligned}$$

Angle measured from the north = $90^\circ - 53.3^\circ = 36.7^\circ$

$$\Delta v = 8.79 \text{ m s}^{-1} \text{ N } 36.7^\circ \text{ W}$$

$$9 \quad \text{a} \quad 40 - 25 = 15 \text{ km h}^{-1}$$

$$\text{b} \quad 25 - (-40) = 25 + 40 = 65 \text{ km h}^{-1} \text{ i.e. } 65 \text{ km h}^{-1} \text{ south}$$

$$\begin{aligned}
 10 \quad v^2 &= (v_2)^2 + (-v_1)^2 \\
 &= (30)^2 + (30)^2 \\
 v &= \sqrt{900 + 900} \\
 &= \sqrt{1800} \\
 &= 42.4 \text{ km h}^{-1}
 \end{aligned}$$

$$\tan \theta = \frac{30}{30}$$

$$\begin{aligned}
 \theta &= \tan^{-1}(1) \\
 &= 45^\circ
 \end{aligned}$$

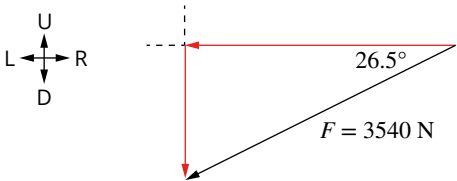
$$v = 42.4 \text{ km h}^{-1} \text{ N } 45^\circ \text{ W}$$

Section 6.4 Vector components

Worked example: Try yourself 6.4.1

CALCULATING THE PERPENDICULAR COMPONENTS OF A FORCE

Use the direction conventions to determine the perpendicular components of a 3540 N force acting on a trolley at a direction of 26.5° down from horizontal to the left.

Thinking	Working
Draw F_L from the tail of the 3540 N force along the horizontal, then draw F_D from the horizontal line to the head of the 3540 N force.	
Calculate the left component of the force F_L using $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$	$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$ $\text{adj} = \text{hyp} \cos \theta$ $F_L = 3540 \times \cos 26.5^\circ$ $= 3168 \text{ N left}$
Calculate the downwards component of the force F_D using $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$	$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$ $\text{opp} = \text{hyp} \sin \theta$ $F_D = 3540 \times \sin 26.5^\circ$ $= 1580 \text{ N downwards}$

Section 6.4 Review

KEY QUESTIONS SOLUTIONS

1 a $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$

$$\text{opp} = \text{hyp} \sin \theta$$

$$F_D = 462 \times \sin 35.0^\circ$$

$$= 265 \text{ N downwards}$$

b $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$

$$\text{adj} = \text{hyp} \cos \theta$$

$$F_R = 462 \times \cos 35.0^\circ$$

$$= 378 \text{ N right}$$

2 $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$

$$\text{adj} = \text{hyp} \cos \theta$$

$$F_S = 25.9 \times \cos 40.0^\circ$$

$$= 19.8 \text{ N south}$$

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\text{opp} = \text{hyp} \sin \theta$$

$$F_E = 25.9 \times \sin 40.0^\circ$$

$$= 16.6 \text{ N east}$$

Therefore, 19.8 N south and 16.6 N east

3 $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$

$$\text{adj} = \text{hyp} \cos \theta$$

$$v_N = 18.3 \times \cos 75.6^\circ$$

$$= 4.55 \text{ m s}^{-1} \text{ north}$$

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\text{opp} = \text{hyp} \sin \theta$$

$$v_W = 18.3 \times \sin 75.6^\circ$$

$$= 17.7 \text{ m s}^{-1} \text{ west}$$

Therefore, 4.55 m s⁻¹ north and 17.7 m s⁻¹ west

4 $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$

$$\text{adj} = \text{hyp} \cos \theta$$

$$s_S = 47.0 \times \cos 66.3^\circ$$

$$= 18.9 \text{ m south}$$

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\text{opp} = \text{hyp} \sin \theta$$

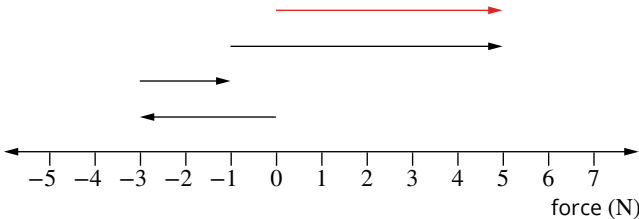
$$s_E = 47.0 \times \sin 66.3^\circ$$

$$= 43.0 \text{ m east}$$

Therefore, the student is 18.9 m south and 43 m east of his starting point.

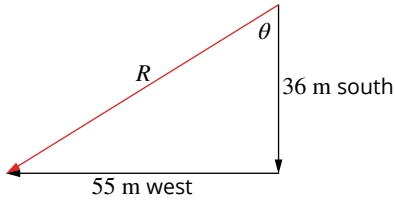
- 5 $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$
 $\text{adj} = \text{hyp} \cos \theta$
 $F_N = 235\,000 \times \cos 62.5^\circ$
 $= 109\,000 \text{ N north}$
 $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$
 $\text{opp} = \text{hyp} \sin \theta$
 $F_W = 235\,000 \times \sin 62.5^\circ$
 $= 208\,000 \text{ N west}$
- 6 **a** $F_S = 100 \cos 60^\circ = 50 \text{ N south}$
 $F_E = 100 \sin 60^\circ = 87 \text{ N east}$
b $F_N = 60 \text{ N north}$
c $F_S = 300 \cos 20^\circ = 282 \text{ N south}$
 $F_E = 300 \sin 20^\circ = 103 \text{ N east}$
d $F_v = 3.0 \times 10^5 \sin 30^\circ = 1.5 \times 10^5 \text{ N up}$
 $F_h = 3.0 \times 10^5 \cos 30^\circ = 2.6 \times 10^5 \text{ N horizontal}$
- 7 horizontal component $F_h = 300 \cos 60^\circ = 150 \text{ N}$
vertical component $F_v = 300 \sin 60^\circ = 260 \text{ N}$
- 8 vertical $= 30.0 \times \sin 50.0^\circ = 23.0 \text{ ms}^{-1}$
horizontal $= 30.0 \times \cos 50.0^\circ = 19.3 \text{ ms}^{-1}$
- 9 Distance south $= 340 \times \sin 45^\circ = 240 \text{ m}$
- 10 Horizontal component of force $= 400 \times \cos 70.0^\circ = 137 \text{ N}$

CHAPTER 6 REVIEW

- B and D are both scalars. These do not require a magnitude and direction to be fully described.
- A and D are vectors. These require a magnitude and direction to be fully described.
- The vector must be drawn as an arrow with its tail at the point of contact between the hand and the ball. The arrow points in the direction of the 'push' of the hand.
- Vector A is drawn twice the length of vector B, so it has twice the magnitude of B.
- Signs are useful in mathematical calculations, as the words north and south cannot be used in an equation.
- 34.0 ms^{-1} north and 12.5 ms^{-1} east. This is because the change in velocity is the final velocity plus the opposite of the initial velocity. The opposite of 34.0 ms^{-1} south is 34.0 ms^{-1} north.
- + 80 N or just 80 N
- 70° down from horizontal to the left or 20° up from vertical to the left
- 

The resultant vector is 5 N right.

- 10 The vectors are $(+45.0) + (-70.5) + (+34.5) + (-30.0)$. This equals -21.0 . Backwards is negative, therefore the answer is 21.0 m backwards.

11


$$R^2 = 36^2 + 55^2$$

$$= 1296 + 3025$$

$$R = \sqrt{4321}$$

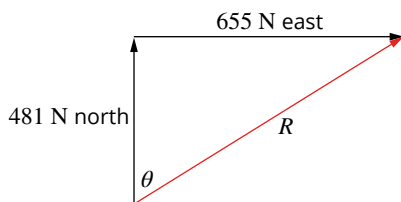
$$= 65.7 \text{ m}$$

$$\tan \theta = 55 \div 36$$

$$\theta = \tan^{-1} 1.5278$$

$$= 56.8^\circ$$

Therefore, the addition of 36 m south and 55 m west gives a resultant vector to three significant figures of 65.7 m S 56.8° W.

12


$$R^2 = 481^2 + 655^2$$

$$= 231\,361 + 429\,025$$

$$R = \sqrt{660\,386}$$

$$= 813 \text{ N}$$

$$\tan \theta = 655 \div 481$$

$$\theta = \tan^{-1} 1.3617$$

$$= 53.7^\circ$$

Therefore, the resultant vector is $R = 813 \text{ N, N } 53.7^\circ \text{ E}$.

13 Taking right as positive:

$$\Delta v = v - u$$

$$= -3 + (-3)$$

$$= -6$$

$$= 6 \text{ ms}^{-1} \text{ left}$$

14
$$\Delta v^2 = (v_2)^2 + (-v_2)^2$$

$$= 18.7^2 + 13.0^2$$

$$\Delta v = \sqrt{349.69 + 169}$$

$$= \sqrt{518.69}$$

$$= 22.8 \text{ ms}^{-1}$$

$$\tan \theta = \frac{18.7}{13.0}$$

$$\theta = \tan^{-1} 1.4385$$

$$= 55.2^\circ$$

$$\Delta v = 22.8 \text{ ms}^{-1} \text{ N } 55.2^\circ \text{ W}$$

$$15 \quad \Delta v^2 = (v_2)^2 + (-v_2)^2 \\ = 55.5^2 + 38.8^2$$

$$\Delta v = \sqrt{3080.25 + 1505.4416} \\ = \sqrt{4585.69} \\ = 67.7 \text{ ms}^{-1}$$

$$\tan \theta = \frac{38.8}{55.5}$$

$$\theta = \tan^{-1} 0.6991 \\ = 35.0^\circ$$

$$\Delta v = 67.7 \text{ ms}^{-1} \text{ N } 35.0^\circ \text{ W}$$

$$16 \quad \sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\text{opp} = \text{hyp} \times \sin \theta$$

$$F_E = 45.5 \times \sin 60.0^\circ \\ = 39.4 \text{ N east}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\text{adj} = \text{hyp} \times \cos \theta$$

$$F_S = 45.5 \times \cos 60.0^\circ \\ = 22.8 \text{ N south}$$

$$17 \quad u = 400 \text{ ms}^{-1}$$

$$\theta = 50.0^\circ$$

$$u_v = u \sin \theta$$

$$= 400 \sin 50.0^\circ \\ = 306 \text{ ms}^{-1}$$

$$18 \quad \text{Findlay horizontal} = 200 \cos 60.0^\circ = 100 \text{ N}$$

$$\text{Dougie horizontal} = 400 \cos 50.0^\circ = 257 \text{ N}$$

$$\text{Total horizontal force} = 357 \text{ N to the right}$$

$$19 \quad \text{Note: 5.0 m distance is not needed}$$

$$\text{Vertical component of velocity} = v \cos 20.0^\circ = 3.00$$

$$v = 3.19 \text{ ms}^{-1}$$

$$20 \quad v = 10.0 \text{ ms}^{-1}$$

$$\theta = 45.0^\circ$$

$$v_v = v \sin \theta = 10.0 \sin 45.0^\circ = 7.07 \text{ ms}^{-1} \text{ down}$$

$$v_h = v \cos \theta = 10.0 \cos 45.0^\circ = 7.07 \text{ ms}^{-1} \text{ to the right}$$